

**INFLUENCE OF HEALTH CARE PROFESSIONALS ON THE UTILIZATION OF
HERBAL MEDICINE AMONG PEDIATRIC PATIENTS IN KISII COUNTY, KENYA.**

BY

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DECLARATION

I declare that this research proposal is my original work and has not been presented in any other university.

Signed.......... Date..... 08.11.2024

This research proposal has been submitted for examination with our approval as university supervisors.

Signed.......... Date..... 25/11/2024
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CONTENTS

DECLARATION	i
ABSTRACT.....	iii
CHAPTER 1: INTRODUCTION	1
Introduction.....	1
Background.....	1
Statement of the problem.....	3
Purpose of the study.....	3
Objectives of the study.....	4
Study Hypotheses.....	4
Research questions.....	4
Study justification	5
Limitation of the study.....	5
Delimitation of the study	6
1.1.9 Assumptions of the study.....	6
1.2.0 Definition of terms.....	6
CHAPTER TWO: LITERATURE REVIEW	7
2.1.0. Introduction.....	7
2.1.1. Literature on topic.....	10
2.1.2 Literature on methods	15
2.1.3. Conceptual framework.....	16
CHAPTER THREE: METHODOLOGY	18
3.1.0 Introduction.....	18
3.1.1 Research design	18
3.1.2. Target population.....	18
3.1.3 Sampling Methods	19
3.1.4 Instrumentation	20
3.1.5 Methods of data collection.....	21
3.1.6 Data analysis methods.....	23
REFERENCES	25
Appendix 1.....	29
Appendix 2.....	30

ABSTRACT

Traditional herbal remedies play a significant role in the health structures of many communities. World Health Organization estimates suggest that nearly 80% of African communities used herbal drugs in 2008. Many prescription medications originated as herbal remedies, however there are many challenges facing utilization of herbal medicines. Health care professionals in settings where traditional therapies are widely used need to gain more knowledge on herbal medicines so that they can be able to assist the community utilize them more efficiently.

The purpose of this to seek to determine the role played by health care professionals in the utilization of herbal remedies among pediatric populations in Kisii county and the factors that may influence the utilization of these remedies.

The design to be used will be a cross-sectional survey.

Data obtained from the study will be analyzed using statistical package for social sciences. The data will be presented in numbers, percentages, median and ranges. Chi-square will be performed to discrete variables to test associations.

CHAPTER 1: INTRODUCTION

Introduction

Herbal medicines have been part of the health care all over the world from the earliest period of existence of mankind, and are still relevant and widely used in Kenya. Precisely, medicinal plants have been almost the sole source of medicaments, either in the form of traditional preparations or as a pure active ingredient [Orwa, J.A, 2002a, p.341].

Traditional medicine has existed along with human civilization. Many societies around the globe have utilized herbal materials as traditional medicine to heal from various diseases [schulz et al.,2004]

In recent years, the importance of utilization of herbal medicine in primary health care system has been emphasized in Africa as well as in other continents. In Kenya, herbs are commonly consumed in various ways, most commonly in form of infusion or decoction prepared from dried plant material [Orwa, J.A, 2002b, p.341]

A lot has shaped the course of herbal medicine in Kenya in the recent decades. However, it faces problems with regulation, low appreciation and acceptance and dwindling plant resources are some of the impediments to development of herbal medicine in Kenya. However, a health care system struggling to cope with demand, high costs and adverse effects of conventional therapy as well as drug resistance have all served to give a lifeline to herbal medicine in Kenya. Moreover, there has been a steady increase in the number of scientific works that contribute to validating therapeutic claims on medicinal plants made by traditional medicine practitioners in Kenya. However, if enormous potential of herbal medicine in Kenya is to be fully realized, calculated steps must be taken towards collaboration in herbal medicine matters [Gakuya et al., 2020]

Background

The World Health Organisation (WHO) estimates that up to 80% of the population in developing countries use Traditional Medicine. In addition, trade in herbal medicine is gaining acceptance globally and is now a lucrative business contributing to the economy. This has been attributed partly due to unavailability of health-care facilities and affordability (WHO,2008,2002).

The practice of herbal of herbal medicine in Kenya has largely been considered primitive by the elite, over the years the practice of herbal medicine has been downgraded as the result of introduction of conventional medicine that is available in a more patient friendly formulation with proper prescription and dosages. In most Kenyan communities the practice of herbal medicine is still considered a family affair and the knowledge passed down along family lines [Kigen et al., 2013]

Herbal medicine user's belief traditional medicine practitioners are more approachable, accessible and their drugs affordable compared to modern medicine. This has led to an ever-increasing demand for traditional medicinal products [Chebii et al.,2020].

Currently because of its importance, herbal medicine is still used and believed by many societies. Societies still believe in the benefit of traditional medicine in human life [Liu et al.,2011].

Despite the increasing use of herbal medicine in many developing and industrialized countries and the reported dependence of the world's population on herbal medicine as their primary form of healthcare, it cannot be fully integrated into the National Health Care scheme in Nigeria, if the orthodox medical doctors are not well disposed towards its use [Awodele O. 2012].

In Kenya, many communities especially from the poor rural areas still rely on herbal remedies. In addition, many Kenyans still believe in the potency of herbal medicine, even when they can access modern medicine. In many cases, they would choose to combine both herbal and modern medicine [Nagata et al.,2011].

Botanical remedies are still popular in treatment if pediatric diseases both internationally and locally. In Germany about 65% of receive one or more herbal medicinal products per year [Nieber et al., 2020]. Nieber, in the PhytoVis project evaluated data set from pediatric population using herbal medicinal products found out that majority of the diseases included common cold, fever, digestive complaints, skin diseases and sleep disturbances.

Many people in Kenya are already taking herbal medicines as self-medication. These are usually prepared at home or obtained from herbalists, pharmacies and supermarkets [Mwangi et al.,2020]. Mwangi further notes that unlike other countries in the world, health care professionals especially medical doctors do not receive any training on herbal medicine in comparison with countries like Germany and France where all health professionals receive training in herbal

medicines which form a core part of their treatment options. In Asia medical doctors acquire post graduate Degrees or Diplomas in herbal medicine.

In Tharaka Nithi county, Kenya, the utilization of herbal medicine among children under five years is as high as 89.4%, [Nzuki & Mwikya, 2016-02a].

A study conducted at Kisii Teaching and Referral Hospital on the prevalence of herbal medicine use in admitted children indicated 62.3% of children had used or were still using herbal medication. [SJO Aula, 2011].

These findings demonstrate that herbal medicinal use is still popular and may increase popularity in use amongst pediatric patients and therefore should not be neglected.

Statement of the problem

In the past decades, much research that has been done on factors influencing utilization of herbal medicine and among those factors include; age of children under-five, education level, occupation, monthly income and location of residence of the caregiver of which all have significant relationship with use of herbal medication, [Nzuki & Mwikya, 2016-02b). Nzioka and Mwika discovered that children under the age of five years get treated more with herbal medicines, lower education level among parents, distance far from hospital and low monthly income favor the use of herbal medicines. However, no research has ever been conducted regarding the role and influence of health care workers in the utilization of herbal medicine among pediatric patients in Kenya.

It therefore remains unclear whether the general training of health care workers, knowledge, attitude and perception towards herbal medicines can have a direct bearing on the consumption of herbal remedies in the community.

Purpose of the study

The study aims to point out the invaluable influence Health Care Professionals have upon parents as pertains the use of herbal medicines on their children during health and illness.

Objectives of the study

A. Broad objective

To assess the influence of health care professionals of Kisii county on the utilization of herbal medicines among pediatric patients.

B. Specific objectives

1. To assess knowledge, attitude and practices towards herbal medications among Health Care Professionals that may influence utilization of herbal medicines among pediatric patients.
2. To determine to what extent herbal medications are utilized by Health Care Professionals and their families among pediatric patients.
3. To determine the main factors that may facilitate or hinder the utilization of herbal medicines among pediatric patients.

Study Hypotheses

H1: Assuming health care professional had enough knowledge of herbal medicines as good as that of conventional medicine, health care professionals would in favor of herbal medicine prescription in pediatric patients.

H2: Professional liability stands out as one of the greatest hindrance in the prescription of herbal medicines by health care professionals.

Research questions

1. Have the health care professionals prescribed any herbal medicines to pediatric patients or for their children?
2. Where did the health care professionals get the knowledge on herbal medicines: experience or academic study? What is their level of knowledge? What is their opinion on enhancement of the knowledge?
3. Are there diseases in children health care professionals feel are only treatable with herbal medicines?

4. For what purpose do health care professionals prescribe herbal medicines: complimentary or alternative medicine?
5. How do health care utilize herbal medicines: personal use or prescription?

Study justification

Herbal medicine use is common among many communities in the world. More parents are considering the use of herbal remedies to maintain their children's good health and to treat their illnesses. They look to pediatricians and other primary health care clinicians for advice concerning the safety and efficacy of herbal products for their children [Woolf AD.,2003].

The training of health care providers varies across different professions, different levels and different countries. This creates variation on knowledge on herbal remedies. However, these variations have not been scientifically proven through research. Its therefore through this research that I seek to address this research gap. The findings of which will be utilized by concerned parties including policy makers in the Health ministry, Education ministry, ministry of culture and as well can be used to lay a basis for further research.

Limitation of the study

This research study faces a few limitations just like many other studies. These limitations are associated the study subjects. Of worth to address is the ethnic background the race of origin and the number of years' experience in practice of the subjects. We all know that the practice of herbal medicine is deeply entrenched and varies across different communities and cultures. This study will generalize the questionnaire to the respondents without consideration of race or ethnicity.

Another challenge will be that of institution of training of the respondents. There are those trained within the country and those trained outside the country. Some countries have incorporated herbal medicine in their training curriculum. This study will not segregate respondents based on this component. The basis will be on licensure to practice within the country health care system.

Delimitation of the study

The study will only investigate on matters to do with pediatric patients and will not involve traditional medical practitioners. It will involve respondents who have formal training from recognized and accredited medical training institutions.

1.1.9 Assumptions of the study

It assumed that the responses given by the respondents are true and are the representative of true findings. Validity and reliability will be addressed through appropriate sample representation and the use of correct measurements.

1.2.0 Definition of terms

1. **Herbal medicine:** WHO defines herbal medicine as plant derived material (that contains either raw or processed ingredients from one or more plants) that provides therapeutic or other human health benefits.
2. **Pediatric:** refers to infants and children under five years
3. **Health care professional:** someone who provides health services to patients, including care for physical or mental conditions. They study physical or mental illnesses and injuries that can occur to humans. They diagnose and treat these health conditions, while some focus on disease prevention.
4. **Traditional practitioner:** a person who practices traditional medicine, incorporating plant, animal, and mineral-based medicines, spiritual therapies, manual techniques and exercise to treat, diagnose, and prevent illnesses or maintain well-being.

CHAPTER TWO: LITERATURE REVIEW

2.1.0. Introduction

Much research work has been done both locally and internationally in regards to herbal remedies. In this literature review I will rely on both local and international research works and articles. However, much of the focus will be on local research work but for themes that may require international research works to develop.

The body has been divided into two parts; the first section contains literature from secondary sources and is purely academic. However there is not much work published in academic books on herbal remedies and therefore relied on internet searches. In this section again is contained a brief history of herbal medicine establishment in Kenya. The second section contains literature on topic from researched articles.

Development and Governance of traditional medicine in Kenya

Governance expresses the organization of people and exercise of power whether formally or informally and the ability to formulate rules on how to attain set goals and objectives at local, social, institutional, national or global levels [Kleeman et al (2006)] 13. Traditional medicine practices can be governed to some extent through local customs and indigenous knowledge normally transferred through cultural means. However, there is need to need to appreciate and harmonize the traditional systems of governance with modern formulated policies, laws and bi-laws passed by county and national governments [Chebii et al] 14.

Modern governance practices are guided and shaped by national laws, county by-laws, act of parliament and policies. The following acts of parliament, laws, policies, bills and gazette notices impact to regulate traditional medicine practice in Kenya [Appendix 1]:

Witchcraft act, 1925, cap 67 laws of Kenya: outlawed any forms of witchcraft practices that were detrimental to the colonial government administration. Traditional medicine practitioners were generally labeled witchdoctors or often charged for practicing witchcraft and being in possession of outlawed charms. The authentic tradition healers lived in constant fear and risked being punished. This act served to slow down the advancement of traditional medicine in the pre-independent Kenya.

Alma Ata declaration, 1978: recognized and acknowledged the roles played by midwives, community Healthcare workers and traditional medicine practitioners in the provision of primary health care.

Development plan, 1989-1993: recognized traditional medicine and set agenda for the promotion of the Traditional medicine practitioners' social welfare and work environment including their registration.

Kenya National Drug Policy, 1994: acknowledged traditional medicine as a key component of Kenya's culture and thus need to mainstream it into the primary health care system.

Registration and recognition of traditional medicine practitioners, ministry of gender, sports, culture and social services, form DC1: this helped to authenticate traditional medicine.

National policy on traditional medicine and regulation of herbal medicine, 2005: this led to establishment of a national coordinating office, an expert committee, clear regulatory framework, national pharmacopoeia, national monograph, registration system and solid manufacturing system.

Sessional paper on traditional medicine in Kenya (2009): anchored five key objectives that promoted traditional medicine namely, regulation, setting up of relevant institutions, contribution of traditional medicine in health care delivery, safety and efficacy and conservation of medicinal plants. The paper highlighted the enforcement of ethical principles in traditional medicine practice which includes equity, fairness and rights to access medical care. It also proposed commercialization of traditional medicine, management of information disclosure and setting up of robust institutions, laws and policies to govern traditional medicine in Kenya.

Pharmacy and poisons board (2010): provided guidelines for submission of traditional herbal medicine and complementary products for registration and licensing.

The traditional medicine and medicinal plants bill, 2010: properly defined traditional medicine and medicinal plants. It proposed for creation of a Traditional Medicine Management Council (TMMC) that was to oversee the practice of traditional medicine in Kenya. TMMC was to draw representation from the Ministry of Agriculture, National Environmental Management Agency (NEMA), Kenya Bureau of Standards (KEBS), Kenya Plant Health Inspectorate

Services (KEPHIS), Kenya Medical Research Institute (KEMRI), National Council of Science and Technology (NCST) and the Kenya Industrial Property Institute (KIPI). The bill also set an eligibility criteria for recognition and certification of traditional medicine practitioners. Eligible candidates are required to have acquired formal knowledge in traditional medicine or have completed relevant training. The bill also proposed punishment for rogue traditional practitioners. This bill proved a hindrance for the TMPs practice because of its requirements.

The health bill, 2012: recognized the role of traditional and complementary medicine in the health care sector. It defined health care professional as an individual with professional training or adequate qualifications for provision of medical services. The bill proposed formation of Health Services Authority (KHSa) of traditional and complementary expert was to be appointed as a member of the authority. The bill empowered the cabinet secretary of health in consultation with KHSa provide regulations for better governance of traditional medicine.

The traditional health practitioners' bill, 2014: provided provisions for training, registration and licensing of the traditional health practitioners. The Traditional Health Practitioner Council of Kenya (THPCK) responsible for regulating its practice.

The health bill, 2015: expressed optimism in the ability of traditional medicine regarding prevention, diagnosis and treatment of diseases. It also expressed the need for sound policies that may help regulate the practice of traditional medicine through department of health or national government. The bill engendered a referral system where traditional medicine practitioners could refer patients to modern health care facilities.

The health act no.21 of 2017: empowered the department of health to provide policies and regulatory institutions that guide the practice of traditional and herbal medicine. The act further provided mechanisms for traditional health practitioners to refer patients to modern health care facilities.

The health laws (amendment) bill, 2018 Kenya gazette supplement no.36, National Assembly bills, no.14 recognized traditional and alternative medicine as a health product.

Traditional and alternative medicine policy draft, 2018, ministry of health provided provisions for mainstreaming Traditional and Alternative Medicine into National Health Care System to boost access to health care for all.

The traditional and alternative health practitioners bill, 2019 provided for training, registration and licensing of traditional and alternative health practitioners and spelt out regulatory and disciplinary guidelines.

The health laws (amendment) act, no.5 of 2019 recognized traditional medicine as a health product.

2.1.1. Literature on topic

Herbal medicine

Definition:

Herbal medicine use also referred to as phytomedicine, botanicals or phytotherapy is the use of medicinal plants or herb-based medicines. They are mostly used as complimentary medicines. Most of these medicines have not undergone rigorous scientific studies of effectiveness, safety or quality. In the United States, they are commonly sold over the counter as food supplements. In some countries, particularly Europe, herbal medicines are more regulated and are available by prescription.

i. Indications.

Herbal medicines are mostly as complimentary medicines. They have been historically used a wide range of acute or chronic conditions, including but not limited to; dementia, cancer, wounds, nausea and vomiting, cardiovascular diseases, attention deficit hyperactivity disorder, diabetes, respiratory conditions such as asthma, colds and cough, joint conditions such as arthritis.

ii. Mechanism of action.

Herbal medicines are thought to affect the body physiology in different ways. They may impact brain chemistry, the immune system and appetite. Some work by preventing cellular damage. All these is achieved through the action of substances in plants called phytochemicals.

iii. Formulation.

The medicines may exist in form of ointments, tea preparations, pill or capsule, and tincture.

iv. Efficacy

Herbal medicine effectiveness has mixed results. The quality may be affected by;

- a. Amount of chemicals in the plant.
- b. Harvesting and storage conditions.
- c. Added ingredients.
- d. Processing
- e. Species and part of the plant used.

v. Risks associated with herbal medicines;

Like any other medicines, though with higher risks, herbal medicines may cause seizures in children mostly due to heavy metal poisoning (lead and mercury), in infants may cause liver damage. Others include: toxicity, increase chances of other diseases, drug interactions and dangerous side effect, reduced effectiveness of other drugs and allergies including anaphylaxis.

vi. Common drug interactions;

May reduce effectiveness of or interact with antidepressants, Aspirin and other NSAIDs, birth control pills, antihypertensives, calcium channel blockers, chemotherapy, immuno-suppressants, statins and warfarin.

vii. Common symptoms of adverse effects

Contact health care professional in case of chest pain, diarrhea, dizziness, syncope, headache, palpitations, nausea and vomiting, dyspnea and skin rash.

Factors that may affect utilization of herbal medicine by health care professionals.

The personal use of and professional recommendation for complementary and alternative medicine has been found to be related to nurses' knowledge and type of training background [Halroyd et al.,2008] 30.

Attitude and perception

The main challenge affecting traditional medicine practice is stigmatization due to poor perceptions and attitudes [Gakuya et al.,2020] 16. Gakuya argues that many members of the public continue to harbor negative perceptions of traditional medicine due to continued negative reporting by mainstream media. He further notes that the government is at cross roads on which ministry to place the practice of traditional medicine, currently it is placed under the ministry of gender, sports and culture as opposed to the ministry of Health.

In Arica, the fight against this stigmatization of traditional medicine is thwarted by some African leaders consciously or unconsciously promoting western-world views over their own [Wekesa, 2020] 5. In 1969, Kenya's first president condemned traditional healers as 'uncivilized cheats' who want to capitalize on other people's resources [Harrington, J.(2018) 17. These attitudes still remain prevalent in contemporary Kenya , particularly among medical professionals [A summary of Kenya's Policies, Industrial Arrangements and Legislation on Traditional Medicine] 18.

In a study conducted at Lagos university Hospital in Nigeria on doctors' attitude towards use of herbal medicine found out that 60% of the participants were disposed to the use of herbal medicine and would encourage their patients to use it. 62% felt that herbal medicine had a positive role in patients' care [Awodele et al., 2012] 26.

In Ghana, clinical nurses have a generally positive attitude towards the use of complementary and alternative medicine [Gyasi et al.,2017] 27.

Knowledge and training

Secrecy of the traditional medicine knowledge or practice poses a major challenge in the advancement of traditional knowledge and research [Hishe et al (2016)]; [Kewassa et al (2015)] [21][22].

Unlike other countries in the world, medical doctors in Kenya do not receive any training on herbal medicine [Mwangi et al] 23. Mwangi observes that the healthcare systems of France and Germany provide a very strong role model in that, doctors and pharmacists receive training in herbal medicine. Herbal medicine is a core part of their treatment options. He further says notes that in the UK, there are many training/research institutions and outlets for complementary medicine.

European union' parliamentary assembly member states have been called upon to promote official recognition of herbal medicine in medical faculties, to encourage its use in hospitals, and to encourage allopathic doctors to study it at university level. Herbal medicine together with other complementary medicines , is practiced in public hospitals in Japan, USA, Germany, Canada, Nigeria, Australia and Norway [WHO Traditional Medicine Strategy, Geneva,2002] 24.

Ghanaian nurses despite positive attitude towards complementary and alternative medicine therapies, do not perceive themselves to have sufficient knowledge of complementary and alternative medicine therapies. Integrating this component of education into the nurses' training curriculum can improve their knowledge and professional practice among them [Gyasi et al.,2017] 27.

A broader depth of knowledge regarding the use of herbal medicine could enable primary healthcare nurses to communicate better with their patients about the risks and benefits of use of herbal medications. The nursing profession should consider introducing an education intervention to enable nurses in their response to patients' questions regarding the use of herbal medicine [Tsele-Tebakang, et al (2022)] 28.

In Australia, medical students express interest in, and the need for complementary medicine education regardless of the opinion of it. This will enable them to respond more effectively concerning drug efficacy and drug interactions [Mackenna et al] 29.

As far as herbal medicines are concerned, physicians are not in a better situation than patients since physicians themselves lack the relevant knowledge and hence they are unable to give informed advice to their patients [Mahmood Hilal & Suswan Hilal (2016)] 31.

Government policies and legislation

By 2018, many African countries had managed to create national laws to govern and regulate traditional medicine, however, Mali, Benin and Ghana are reported to have made efforts to integrating them into national health services [Harrington, J.(2028)] 17.

Kenya now has an extensive policy framework and institutional arrangements on traditional medicine. It also has an extensive corresponding several pieces of legislation scattered over several sectors and line ministries touching traditional medicine, however, these extensive arrangement cannot impact positively on traditional medicine because of lack of coordination and harmonization in Kenya's policies , institutional arrangements and legislation to ensure effective management and development of traditional medicine[A summary of Kenya's Policies, Industrial Arrangements and Legislation on Traditional Medicine] 18.

Several challenges face Kenya in integration of Traditional herbal medicine into national health care. These include; lack of national policy and regulatory framework; issues pertaining to safety, efficacy, quality, access and rational use of traditional herbal medicine and lack of cooperation between complementary medicine practitioners and medical practitioners [Mwangi et al.] 23. Mwangi continues to say that lack of clear policy on intellectual property rights and adequate benefit sharing relating to herbal medicine, traditional/indigenous knowledge, biopiracy and unsustainable use of medicinal plants are other challenges.

Lack of proper policy guidelines on cooperation between traditional practitioners and professional medical practitioners creates fear of professional liability in cases of adverse events on use of herbal medicinal products prescribed by professional medical practitioner [Mahmood Hilal & Suswan Hilal (2016)] 31.

Safety, efficacy, and quality of products of traditional medicine practice

There have been made many unsubstantiated claims on efficacy of medicinal plants, concerning the diseases they cure or the nutritive value they provide [Gakuya et al.,2020]. Gakuya notes that the safety and quality of most products borne from traditional medicine are largely unverified. Lack of scientific data to verify the efficacy and safety may lead to wastage of resources on non-efficacious products, which may result to overdoses, side-effects or even death due to ingestion of poisonous material. He continues to say that the competence of the practitioners of traditional medicine is also a cause of concern given the upsurge of the number of practitioners falsely

claiming to have all kinds of cures. There are also a lot of quacks who have entered the traditional medicine market either as practitioners or traders.

In eastern countries herbal medicine is very advanced and widespread. Good laboratory practice and good clinical practice has led to the development of safe and high quality and efficacious medicines from herbal medicines [Gao Wen-Yuan (2005)] 25. Gao further notes that many traditional medicines consisting of mainly herbs have been included in Chinese pharmacopoeia with numerous hospitals, universities and large pharmaceutical industries dealing with herbal medicines.

Presence of unprecedented adverse events and side effects prove a hindrance to the use of traditional herbal medicines [Mahmood Hilal & Suswan Hilal (2016)] 31. In developed countries popularity of herbal medicines has been fueled by concern about the adverse effects of conventional drugs questioning the approaches and assumptions of conventional medicine [Mwangi et al.] 23.

Availability of herbal medicines

In many African countries, accessibility and affordability is the driving force for increased use of herbal medicines [Mwangi et al.] 23.

A health care system struggling to cope with demand, high costs and adverse effects of conventional medicine has served to give a lifeline to traditional medicine [Gakuya et al., 2020] 16.

2.1.2 Literature on methods

In reviewing the literature on the methods, a few notable research works come to fore whose topics are similar to the topic in my proposal. A few are discussed in this section.

Ameade et al.,2015 employs a cross-sectional study design method with a questionnaire to assess the knowledge and attitudes of medical students in Tamale, Ghana towards herbal medicines. He analyzed the data obtained using statistical package for social sciences (SPSS). Cronbach's alpha value was used to assess the consistency and the Pearson's correlation coefficient used to calculate the relationships.

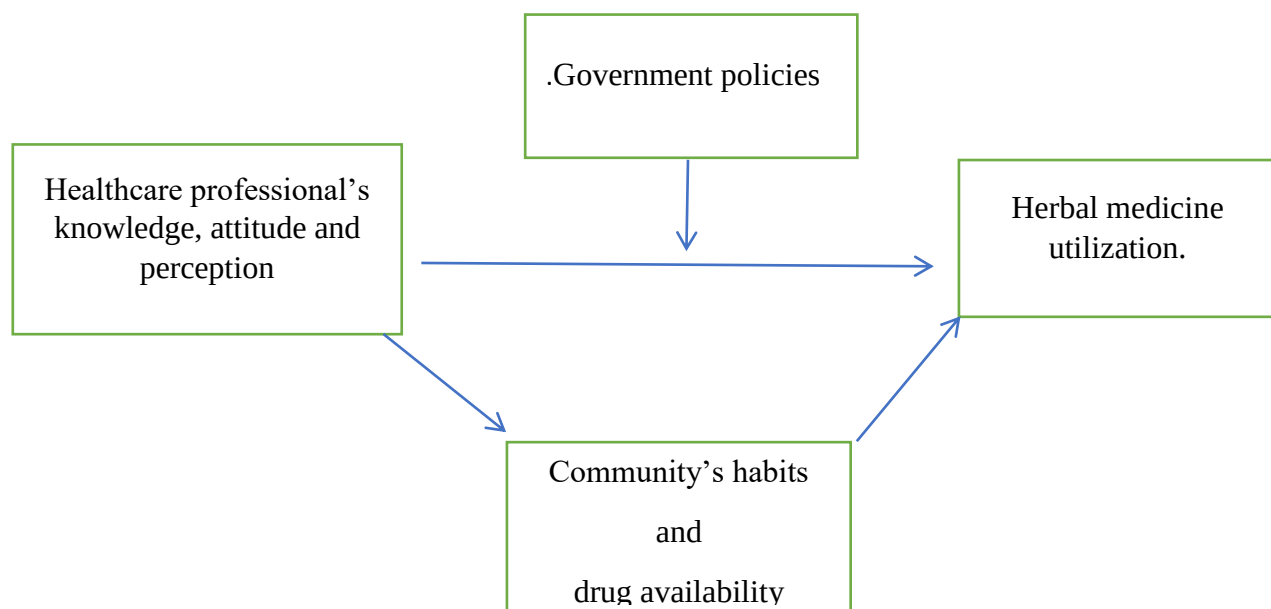
Awodele et al.,2012 in establishing doctor's attitudes towards the use of herbal medicine in Lagos and elucidating the possible factors that may limit their interest in use of herbal medicine, a descriptive cross-sectional survey was conducted using a WHO standard questionnaire to obtain data. The obtained data was analyzed using SPSS.

James and Bach, 2014 in the study of awareness, use, attitude and perceived need for complementary and alternative medicine education among students use a descriptive cross-sectional study using a structured questionnaire . Chi-square, fishers' exact test and general liners model analysis were used to compare data between independent cohorts. Data from the questionnaire enclosed into SPSS.

Ashraf et al.,2019 in assessing and comparing the knowledge, attitude and perception about complementary and alternative medicine between pharmacy and non-pharmacy students, a descriptive cross-sectional study was conducted. A self-administered questionnaire was for data collection. The data was analyzed using an SPSS.

To assess the physicians' knowledge and attitudes toward herbal medicines, a cross-sectional study was conducted (Mahmood Hilal, Sawsan Hilal, 2016). The researchers developed a self-administered questionnaire specifically designed to address the study objectives.

2.1.3. Conceptual framework



The healthcare professional's knowledge on and attitudes towards herbal medication (independent variables) influence the utilization of herbal medicine (dependent variable). The government policies and availability of herbal medicine stand as mediating variables. The community's habits are confounding variables.

CHAPTER THREE: METHODOLOGY

3.1.0 Introduction

In this chapter we look at the design of the research, the target population, sampling technique, the preparation of data collecting instruments, data collection procedures and methods of data analysis.

3.1.1 Research design

The research design for this proposed research will be a cross-sectional descriptive survey. This research design will be sufficient enough to answer my research questions. Because of the short time available to conduct the research, this design will be the best for it will allow me to gather data from a large number of subjects within a short point in time. Besides, it is cheap to conduct the research owing to the researcher being a student with limited financial. The tools for collecting data will be easily developed and administered therefore cutting on costs of time and finances. The design will also allow for identification and studying of correlations.

3.1.2. Target population

The study will be conducted among specific target of health care professionals in Kisii county. The health care professionals include nurses, clinical officers and medical officers. These officers are spread across public health facilities in the nine sub-counties making up Kisii county. These cadres of health care professionals has been chosen because they are the ones mandated to offer consultation services at different levels of care in the Kenyan public health care system. Nurses are the main health care providers at level two health facilities in Kenyan health care system. They are however found across all level of health care levels offering other preventive and promotive health services. Their role in offering care in maternal child health clinics makes them the first contact people to offer advice on various health issues affecting the mother and child including matters to do with herbal medicines as alternative and complimentary treatments. Clinical officers on the other hand are main service providers in the level three and four public health facilities. In level three health facilities are dominated by nurses and clinical officers. Nurses giving services at the maternal child health clinics while clinical officers give their

services at the main outpatient consultation rooms where they make highest and final decisions concerning patient management at this level of care .

In level four public health facilities are also found Nurses and clinical officers in addition to medical officers. Nurses and clinical officers perform the same roles as those in level three facilities. However, the availability of medical officers and specialist doctors makes them the highest decision makers on matters of patient care.

Level five hospitals are the highest in most counties including Kisii county. At this level also are found nurses, clinical officers, medical officers and specialist doctors. They all get in contact with pediatric patients at either in- patient or out -patient levels. Clinical officers being mostly at the out-patient level while the other professionals being at the in-patient levels.

The kisii county public health sector is composed of 900 nurses, 310 clinical officers, 80 medical officers and 4 pediatric consultants giving a target population of 1294 health care professionals. These target population have to be employees of the county government and offering services in the public health facilities.

3.1.3 Sampling Methods

Convenience sampling method will be of choice. Participants will be chosen primarily on availability and willingness to participate in the survey.

Known target population size formula will be used to calculate the sample size:

$$\frac{Z^2 \times p(1-p)}{e^2} \\ 1 + \frac{Z^2 \times p(1-p)}{e^2 \times n}$$

- Z= z square =1.96
- N= population size=1294
- e= confidence interval =0.05

- $p = \text{standard deviation} = 0.5$

Using the above method and substituting the variables, the sample size will be 296 participants.

We now calculate the sample size for every cadre of the three professions;

- nurses;

$900/1294 \times 100 = 70\%$ (the percentage representation of nurses in the target population)

$70\% \text{ of } 296 = 207$

We therefore shall recruit 207 nurses for the study.

- Clinical officers ;

$310/1294 \times 100 = 24\%$ (the % representation of clinical officers in the target population)

$24/100 \times 294 = 71$

We therefore shall recruit 71 clinical officers for the study.

- Medical officers;

$80/1294 \times 100 = 6\%$

$6/100 \times 294 = 18$

We therefore shall recruit 18 medical officers for the study.

Pediatric consultants consist of 0.3 % of the target population therefore will give insignificant contribution to the study. We therefore exclude this cadre.

3.1.4 Instrumentation

The researcher will develop a self-administered questionnaire structured specifically to meet the specific objectives of the study. This will be developed following extensive literature review (Ameade et al.,2015 , Awodele et al.,2012 , James and Bach, 2014, Asharaf et al.,2019 , Mahmood Hilal and Sawsan Hilal.,2016) and shall greatly borrow from these sources.

The developed questionnaire will have a cover page that will contain the purpose of the study and also instructions to the respondents on how to respond to the questions. The main body of the questionnaire will have four parts;

Section one will mainly concern with the health care provider's profile including gender(male/female) cadre(nurse/clinical officer/medical officer), level of qualification(certificate/ diploma/bachelor/master/doctorate), experience(<5 years/5-10years/>10years) and whether they are parents or guardians .

Section two will cover questions about knowledge level (basic/moderate/advanced), knowledge source(experience/academic study/others) and the desire to study more knowledge concerning herbal medicines(yes/no) responses. There will be questions to assess the knowledge of the participant. Attitude will be measured by a five-point likert scale(highly agree/agree/no comment/disagree/highly disagree) consisting of five questions.

Section three will contain a 5-point likert scale questions that evaluate the health care professional's utilization of herbal medicine (personal verses prescription), (complimentary verses alternative) in a scale (never/rarely/sometimes/often/never).

Section four will consist of YES/NO questions that explore the facilitating and hindering factors of herbal medicine utilization.

The questionnaires will be three hundred and twenty in number. Two hundred and ninety four will be distributed in the actual data collection while the remaining will be used as piloting questionnaires.

The questionnaire will be piloted among health care professionals from selected health facilities. The feedback received will be incorporated in the final questionnaire to enhance validity before embarking on the actual fieldwork. Besides that, validity and reliability will be enhanced by using established and validated methods and techniques to collect and analyze data. Data will be collected from adequate sample size. Results will be interpreted and reported accurately and transparently.

3.1.5 Methods of data collection

I. Procedure and authorization;

Before embarking on the mission of data collection, first I have to seek authorization from the county health research committee. The approval and recommendation from the research committee will be done through writing and I will be provided with a written authorization

document. This document will be the one that I will present to health institutions where actual data will be collected.

The data collection area-Kisii county health facilities, will be divided into nine zones. Each zone to represent the nine sub-counties that form the larger Kisii county. The main facilities to be targeted will be the top most establishments within the sub-county, the sub-county hospitals. Then next line will be health centers. These facilities will have been chosen because of the high percentage of health care professionals deployed in these facilities. This will also cut costs for travel involved in the data collection.

At the facility level, I will first seek for the facility administrator to whom I will present the necessary authorization documents. Then I will explain the purpose of my work, the target for the target persons for the research and the overall timelines for the research. The administrator will in turn be required to disseminate this information to the in-line departmental in charges for the three cadres involved (nurses, clinical officers and medical officers). These in charges will also in turn communicate to their respective junior officers. The hospital administrator will be required to do an overall formal communication to the staff informing them of the exercise through a memo to be distributed to all hospital departments.

The questionnaires containing full instructions about how they should be applied and used will be left with the administrators who will be required to distribute them to the respondents. This will help in observing confidentiality. The questionnaire will be collected from respondents after three weeks, this period will be considered sufficient allow for all the respondents willing to take part to have a chance to participate. This period takes into consideration health care professionals who would be a way from duty for various reasons including day offs or night offs.

I, the researcher will however be required maintain contact with the focal person to offer any clarifications on issues arising from the questionnaire administration. I will provide my phone contacts to the participants for the purpose.

II. How to ensure high response rates;

High rate of response will be sought in several strategies. First the questionnaire will be simplified in language and instructions to the understanding of the respondents. Secondly the purpose of the study will be explained to the respondents so that they may understand the nature, reason and application for the data to be collected. Third, the time period offered to the

respondents will long enough to allow them respond at their free time , that they should not feel pressured to do give their responses. Lastly, giving out of phone contacts to the respondents so that issues and concerns and arising from the questionnaire could be addressed in time.

III. Ethical values;

1. Consent ;

The respondents will be made to understand the participation in the exercise will be voluntary and that willing to respond to the questionnaire will be a demonstration of consent.

2. Confidentiality;

The participants will not be allowed to indicate their personal identification details including names, national identification numbers/ personal numbers or their phone numbers. That even if the researcher identifies the respondents by accident, he won't share it the details with anyone.

3. Feedback ;

That the findings from the research will be published and be made available to the participants.

4. Research approval;

- Approval by the university's ethics and review committee.
- Approval by the NACOSTI.

3.1.6 Data analysis methods

Data from completed survey is to be entered into the statistical package for social sciences software.

Frequencies and proportions will be calculated for categorical variables.

Median and interquartile ranges will be conducted for knowledge, attitudes and practical scores.

Chi-square and fishers exact testis to be used to analyze the bivariate relationship between knowledge and attitude and the effect on utilization of herbal medicine.

Spearman's correlation coefficient for knowledge and attitude scores to determine the strength of association between those variables.

p-value of < 0.05 will be considered significant in the analysis .

REFERENCES

1. Awodele O. (2012). Doctor's attitudes towards use of herbal medicine in Lagos, Nigeria. J. Herb. Med.2, 16-22
2. Daniel Waweru Gakuya., Mitchel Otieno Okumu., Stephen Gitahi Kiama., James Mucuru Mbarika., Peter Karuri Gathumbi., Peter Mbaabu Mathiu., Joseph Mwanzia Nguta, (2020). Traditional Medicine in Kenya : Past and Current Status, Challenges, and The Way Forward. <http://creativecommons.org/licenses/by/4.01>
3. [African Journal of Pharmacology and Therapeutics Vol.2 No.1 Pages 32-37,2013](#)
4. Mwangi et al. Traditional Herbal Medicine In Kenya. East and central African Journal of pharmaceutical sciences vol.812.
5. Nagata JM, Jew AR, Kimeu JM, Salmen CR, bukusi EA and Cohen CR [2011], Medical Pluralism on Mfangano Island; use of medical plants among patients living with HIV/AIDS in Suba District, Kenya. J. Ethnopharmacol. 135:501-609.
6. Nzuki, David Mwikya.,(2016). Utilization of Herbal Medicine Among Children Under 5 Years of Age in Tharaka Nithi County, Kenya. <http://ir-library.ku.ac.ke/handle/123456789/15093>
7. Orwa, J.A. (2002). Herbal Medicine in Kenya: Evidence of Safety and Efficacy. East African Medical Journal

8. SJO Aula, (2011) . Traditional Medicine use Among Children Aged 0-12 Years Admitted To Kisii Level 5 Hospital. Erepository.unbi.ac.ke/bitstream/handle/11295/25570

9. Willy Kibet Chebii, John Kaunga Muthee and Karagu Kiemo: Journal of Ethnobiology and Ethnomedicine <https://doi.org/10.1186/s13002-020-00389-x>

10. WHO [2008] Traditional Medicine, fact sheet:<http://www.who.int/mediacentre/factsheets/fs134/en/>.

11. WHO (2002) Traditional Medicine Strategy 2002-2005, WHO Geneva 2002:<http://www.who.int/topics/traditionalmedicine/en/>.

12. Woolf AD.,2003. Herbal remedies and children: Do they work? Are they harmful?PMID:1237916

13. Stoll-Kleeman, S., Bender, S., Berghofer, A., Bertzy, M., Fritz-Yietta, N., Schliep, R. Linking governance and management perspectives with conservation success in protected areas and biosphere reserves. Perspectives on Biodiversity Governance and Management.2006; 1:40

14. Chebii, K. W., Kiemo, K., Muthee, K. J.: The Governance of Traditional Medicine and Herbal Remedies in Selected Markets of Western Kenya <https://doi.org/10.1186/s13002-020-00389-x>.

15. Afr J Tradit Complement Allern Med.2011;8(5suppl):115-123 Trends and Challenges of Traditional Medicine in Africa.doi10.4314/astcam.v8i5s.5

16. Gakuya, D. M., Okumu, M.O., Kiama, S.G., Mbathia, J.M., Gathumba, P.K & Ngutha, J.M (2020) Traditional Medicine in Kenya: Past and Present Status, Challenges, and The Way Forward. <https://doi/10.1016/5.sciaf.2020.e00360>

17. Harrington, J. (2018). Governing Traditional Medicine in Kenya: Problemization and the Role of Constitution. African studies.77,223-239. <https://doi.org/10.1080/00020184.2018.1452856>

18. A summary of Kenya's Policies, Institutional Arrangements and Legislation on Traditional Medicine. Doi:1031014/aor.1996.01.03.23

19. Africa Research Institute Kenya: Traditional Medicine and The Law. www.africaresearchinstitute.org

20. Sifuna, N (2022) Configuring a Model Framework, Statute on Traditional Medicine for Kenya To Be Called ' The Traditional Medicine Coordination Act'. Beijing Law Review,13,183-203.doi:104236/b/r.2022.131012

21. Hishe, M., Asfaw, Z & Giday, M. Review on value chain analysis of medicinal plants and associated challenges. J Med plants studies.2016;4(3):45-55

22. Kewassa, G., Abebe, T & Demissie, A. Indigenous knowledge on the use and management of medicinal trees and shrubs in Dale district, Sidama zone. Ethnobiology Research and Applications; southern Ethiopia; 2015.

23. J. w. Mwangi, N. N. Mungai, G. N. Thothi & I. O. Kibwage, East and Central African Journal of Pharmaceutical Sciences vol.8(2)22-26.

24. WHO Traditional Medicine Strategy 2002-2015, Geneva, 2002

25. Gao Wen-Yuan (2005) personal communication. Department of National products and Traditional Chinese Medicine. Tianjan University of Traditional Chinese Medicine, China.
26. Awodele, O., Agbaje, E.O., Abiola, O. O. & Dalapo D. C. Doctor's Attitudes Towards The Use of Herbal Medicine in Lagos, Nigeria. Journal of Herbal Medicine vol.2. <https://doi.1016/j.hermed.2012.02.002>
27. Abass, K., Accam, B. T., Gyamfi, S. A. & Gyasi, R. M: Nurses Knowledge, Clinical Practice and Attitude Towards Unconventional Medicine: implications for intercultural healthcare. <https://doi.org/10.1016/j.ctp.2017.07.001>
28. Morris-Eyton, H., Pretorious, E. & Tsele-Tebakang, T. Herb-drug interactions: Perceptio and Revelations of nurses in primary healthcare clinics, South Africa. International Journal of African Nursing Sciences vol.16. <https://doi.org/10.1016/j.ijans.2022.100409>
29. Mckenna, L., Robinson, A. & Templeman, K. Student Identification of The Need For Complimentary Medicine Education In Australia Medical Curricula: A construct grounded theory approach. Complimentary therapies in medicine vol.23. <https://doi.org/1016/j.ctim.2015.02.002>
30. Anthony, L. Z., Charlie, C. L., Holroyd, E. & Lerna, K. P. Beliefs and Attitudes Towards Complimentary Medicine Among Registered Nurses in Hong Kong. <https://doi.org/10.1016/j.u.ijnurstu.2008.04.003>
31. Mahmood H & Suwsan H. Knowledge, Attitude and Utilization of Herbal Medicines by Physicians in The Kingdom Of Bahrain; <https://doi.org/10.1016/j.juubas.2016.11.001>
32. Evans P.K Ameade, Anthony Amalba, Gideon K. Helegbe and Baba S. Mohammed. Herbal Medicine: A Survey on the Knowledge and Attitudes of Medical Students in Tamale, Ghana.
33. Ashraf, M., Saeed, H., Saleem, Z . BMC complement Altern Med 19,95(2019).<https://doi.org/10.1186/s12906-019-2503-y>
34. Peter B James & Abdulai J Bah 2014.

Appendix 1

Development plan, Kenya (1989-1993).

Health bill 2015 <http://kenyalaw.org/kl/fileadmin/pdfdownloads/bills/2015/HealthBill2015pdf>

Health laws (Amendment) Act, No 5 of 2019
http://Kenyalaw.org/kl/fileadmin/pdfdownloads/AmendmentActs/2019/HealthLaws_AmmendmentACTNo5of2019.pdf
https://www.researchgate.net/publication/323267945_Acritical_Overview_of_the_Health_Act_2017

Ministry of Gender, Sports, Culture and Social Services, Form DCI, 2003

National policy on traditional medicine and regulation of herbal medicines. Report of a WHO global survey, may 2005

The Alma Ata Declaration https://www.who.int/publications/almaata_declaration_en.pdf

The Health Act No.21 of 2017
<http://kenyalaw.org/kl/fileadmin/pdfdownloads/Acts/HealthActNo.21of2017.pdf>

The Health Laws (Amendment) Bill, 2018, Kenya Gazette Supplement No. 36, National Assembly Bills, No. 14
http://Kenyalaw.org/kl/fileadmin/pdfdownloads/bills/2018/HealthLaws_Amendment_Bill2018.pdf

The Traditional Health Practitioners Bill 2019

The Traditional Health Practitioners Bill 92014)

The Traditional Medicine and Medicinal Plants Bill (2010)

Appendix 2
CONSENT FORM

TITLE: Influence of health care professionals on utilization of herbal medicines among pediatric patients in Kisii county, Kenya.

INVESTIGATOR: Stanley Omariba Siro [HB203/S/17778/22-KYU]

SUPERVISORS: Dr. Wanjiku Njoroge and Dr. Godwill Munyekenye.

INVESTIGATOR NOTE: Thank you for agreeing to read this form. It offers important information about this study which will help you make a decision if you will participate in the study.

INTRODUCTION: herbal medicines play an important role in the health of children in many communities of this country. It is estimated that over 70% of children receive herbal medicine in the course their lives for different reasons. In a community where herbal medicine use is common among pediatric population, it is necessary to understand the health care professionals' knowledge and attitudes towards these remedies and their contribution to the utilization of these medicines.

PROCEDURE: if you agree to participate in this study, you will respond to some questions on the questionnaire provided. You will read instructions on how to respond to the questions before you respond.

CONFIDENTIALITY: The information and responses that you will give will be held in strict confidence and will be used for the purpose of the study.

ETHICAL CONSIDERATIONS: Approval for conducting the study has been granted by the research and ethics committees.

To indicate that you understand the conditions of the study and that you consent to participate, please sign in the spaces provided:

Participant's signature..... Date.....

Investigator's signature..... Date.....

Appendix 3
QUESTIONNAIRE

SECTION 1.

GENDER: Male ☐ Female ☐

CADRE ☐ Nurse ☐ Clinical Officer ☐ Doctor

QUALIFICATION: ☐ Certificate ☐ Diploma ☐ Bachelors' degree

☐ Masters' degree ☐ PhD

WORK EXPERIENCE: ☐ <5 years ☐ 5-10 years ☐ >10 years

PARENTHOOD: ☐ Parent/Guardian ☐ Not parent/Guardian

SECTION 2: Choose either YES or NO in response to the following statements;

1. Herbal medicines are defined as active ingredients of aerial or underground parts of plant or other plant material or a combination thereof whether in crude state or as plant preparations: YES ☐ NO ☐

2. Herbal medicines which forms the basis of Healthcare in the world since earliest days of mankind are still used in Kenya for the treatment of pediatric diseases:

YES ☐ NO ☐

3. In most communities, over 60% of children receive one or more herbal medicine from traditional herbalists per year: YES ☐ NO ☐

4. Concerning the relationship between Age and herbal medicine use: younger children [<5 years] are more likely to be treated with herbal medicines than older children [> 5 years]: YES ☐ NO. ☐

5. Most herbal medicines used for treatment of children have not undergone rigorous scientific studies on effectiveness, safety or quality: YES ☐ NO ☐

6. Would you please give two examples of herbal medicines used for treatment of in children and the diseases they are used to treat:

1.
.....
2.
.....

7. Herbal medicines used in children are always safe and will not cause drug interactions when used with other modern (conventional) medicines:

YES ☐ NO ☐

8. In infants and children signs of herbal medicine toxicity may include;

1. Reduced level of consciousness and seizures; YES ☐ NO ☐

2. Yellowing of the eyes; YES ☐ NO ☐

Tick as appropriate ;

A) What is the source of your knowledge on herbal medicines?

☐ Academic study

☐ Experience

☐ Others: [community members, inheritance]

B) How would you rank your level of knowledge on herbal medicines?

☐ Basic

☐ Average

☐ Advanced

SECTION 3: choose the most appropriate response from the five options given to each statement.

1. Have you used herbal medicine for your children or for other children under your care?

☐ Always

☐ Often

☐ Sometimes

☐ Rarely

☐ Never

2. Have you or do you prescribe herbal medicines to other pediatric patients?

☐ Always

☐ Often

☐ Sometimes

☐ Rarely

☐ Never

3. Do you use herbal medicines a substitute to conventional (modern) medicine?

☐ Always

☐ Often

☐ Sometimes

☐ Rarely

☐ Never

4. Do you use herbal medicines together with other conventional (modern) medicines?

☐ Always

☐ Often

☐ Sometimes

☐ Rarely

☐ Never

5. Pediatric patients whose HEALTH CARE PROFESSIONAL knows about herbal medicine in addition to conventional (modern) medicine benefit more than those whose HEALTH CARE PROFESSIONALS know of modern medicine only.

☐ Strongly agree

- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly disagree

6. Knowledge about herbal medicine should be included in medical curriculum for medical students so that all health care professionals can have enhanced knowledge.

- ☐ Strongly agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly disagree

7. HEALTH CARE PROFESSIONALS should recommend the use of herbal medicine in the community.

- ☐ Strongly agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly disagree

8. Herbal medicine should be integrated with modern medicine in health care system to improve health care coverage.

- ☐ Strongly agree
- ☐ Agree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly disagree

9. Traditional herbal medicines can cure some disease in children that modern medicine cannot cure.

- ☐ Strongly agree

- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly disagree

10. Caregivers should be given information about the role of herbal medicine in the health of children.

- ☐ Strongly agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly disagree.

SECTION 4: Tick where appropriate.

The following factors influence the utilization of herbal remedies by health care professionals. Choose by ticking whether they are facilitating or hindering factors.

FACTOR	FACILITATING	HINDERING
1.Low cost of herbal medicine		
2. Lack of sufficient knowledge on herbal medicine		
3.Ineffectiveness of modern (conventional) medicine		
4.Lack of availability of herbal medicine		
5.Fear of professional liability on using herbal medicine		
6.Poor quality control on herbal medicines		
7.Increased side effects and adverse events of modern(conventional) medicine		
8.Lack of formal regulation of herbal medicine by the government		
9.Limited information on adverse events of herbal medicine		
10.Lack of integration of herbal medicines and modern medicines in health care systems .		

Appendix 4

BUDGET

The main costs for the project will include but not limited to;

1. Printing of 300 copies of the consent form.
2. Printing of 300 copies of the questionnaire.
3. Travel and networking costs.
4. Technological costs associated with data analysis.

ITEM	COST
1.Printing of 300 copies of consent forms @ KSH.10	3,000
2.Printing of 300 copies of questionnaires @ KSH.50	15,000
3.Travel and networking costs	15,000
4.Technological costs associated with data analysis software	50,000
TOTAL	83,000

Appendix 5

Map

Kisii County

